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CMSC 207 CRN 32950

CMSC 207: Chapter 8 Homework

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1. Let R be a relation defined on the set of all integers $\mathbb{Z}$ as follows:

$$\text{for all integers } m \text{ and n}, m\,R\,n \iff 5 \mid (m-n).$$

Prove that R is an equivalence relation on $\mathbb{Z}$.

Let m, n, and p be arbitrary integers.

Proof: R is reflexive.

Since $5 \mid (m-m)$ is equivalent to the statement $5 \mid 0$, which is true, $m\,R\,m$, so R is reflexive.

Proof: R is symmetric.

Suppose $m\,R\,n$. By the definition of R, we have $5 \mid (m-n)$. This means that there exists some integer k such that $m-n=5k$. By elementary algebra:

$$m-n=5k \to n-m=5(-k).$$

Since $-k$ is an integer, by the definition of divisibility, $5 \mid (n-m)$. By the definition of R, we have $n\,R\,m$. Therefore, R is symmetric.

Proof: R is transitive.

Suppose $m\,R\,n$ and $n\,R\,p$. By the definition of R, we have $5 \mid (m-n)$ and $5 \mid (n-p)$. This means that there exist some integers k_1 and k_2 such that $m-n=5k_1$ and $n-p=5k_2$. By elementary algebra:

$$m-n=5k_1 \to m=n+5k_1.$$
$$n-p=5k_2 \to n=p+5k_2.$$

Substituting the second equation into the first gives:

$$m=\left(p+5k_2\right)+5k_1=p+5\left(k_1+k_2\right).$$
$$m-p=5\left(k_1+k_2\right).$$

Since k_1+k_2 is an integer, by the definition of divisibility, $5 \mid (m-p)$. By the definition of R, we have $m\,R\,p$. Therefore, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on $\mathbb{Z}$. ∎

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2. Let S be the set of all strings of 0s and 1s of length 3. Define a relation R on S as follows:

$$\text{for all strings } s \text{ and } t \in S, s\,R\,t \iff \text{the two left-most characters of } s \text{ and } t \text{ are the same.}$$

Prove that R is an equivalence relation on S.

S can be expanded as follows:

$$S = \{000, 001, 010, 011, 100, 101, 110, 111\}.$$

By the definition of R, we can partition S into the following equivalence classes:

$$[000] = \{000, 001\}.$$
$$[010] = \{010, 011\}.$$
$$[100] = \{100, 101\}.$$
$$[110] = \{110, 111\}.$$

Let s, t, and u be arbitrary strings in S.

Proof: R is reflexive.

Since the two left-most characters of s are the same as themselves, $s\,R\,s$, so R is reflexive.